

25.09.2024 CSE211 - FUNDAMENTALS -

Definition (sets): A set is any well-defined collection of objects called the elements or members

capital $A = \{a, b, c\}$ $\rightarrow$ finite $\mathbb{Q} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0 \right\}$

$\mathbb{R} \rightarrow$ real numbers

$\mathbb{Q} \rightarrow$ rational numbers

$\mathbb{Z} \rightarrow$ integer

$$\mathbb{N} = \{1, 2, \dots\} \rightarrow \text{infinite}$$

subset: If every element of A is also an element of B , then A is subset of B .

$$\forall a \in A \Rightarrow a \in B$$

$$\forall a \in A \text{ and } a \in B \Rightarrow A \subset B$$

if A is not a subset of B then $A \not\subset B$

empty set: Empty set is a set with no element

$\{ \}$ or $\emptyset \rightarrow$ notation

exmpl: $A = \{ x \mid x \in \mathbb{Z} \text{ and } x^2 < 12 \}$

$B = \{ x \mid x \in \mathbb{Z} \text{ and } x^2 < 0 \}$

$= \emptyset$

$A = \{ -3, -2, -1, 0, 1, 2, 3 \}$

Cardinality: The cardinality of a set is the number of elements in the set

* A set A is called finite if it has n distinct elements, where $n \in \mathbb{N}$

$|A| = n \rightarrow$ notation

* A set A which is not finite is called infinite.

$\rightarrow$ künnel kümes;

Power Set: If A is a set, then the set of all subsets is called "the power set of A "

$P(A) \rightarrow$ notation exmpl: $A = \{ a, b, c \}$

$P(A) = ?$

$P(A) = \{ \{ a \}, \{ b \}, \{ c \}, \{ a, b \}$

$\{ a, c \}, \{ b, c \}, \{ a, b, c \}, \{ \} \}$ $\rightarrow \binom{|A|}{0} + \binom{|A|}{1} + \binom{|A|}{2} + \binom{|A|}{3}$
(sonra gelicez)

Q2-) if $|A| = n$, $|P(A)| = ?$

$|P(A)| = 2^n$

Note: The empty set is a subset of any set.
Every set is its subset.

$A \subseteq A, A \supseteq A, A = A$

Operations on Sets

α If A and B are sets, we define their union as the set containing of all elements that belong to A or B.

$$A \cup B$$

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

} notation

α If A and B are sets, we define their intersection as the set consisting of all elements that belong to both A and B.

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

α If A and B are two sets, the complement of B with respect to A as the set of all elements that belong to A but not B.

$$A - B \text{ or } A \setminus B$$

$$A - B = \{x \mid x \in A \text{ and } x \notin B\}$$

α The universal set is a set that contains all the sets.

U → notation

α If U is a universal set containing A, then $U - A$ is called its complement

$$A' = U - A \text{ or } \bar{A}$$

$$\bar{A} = \{x \mid x \in A\}$$

-RABIS

Algebraic Properties of Set Operations

Let A and B sets,

Commutative properties:

$$\left. \begin{array}{l} A \cup B = B \cup A \\ A \cap B = B \cap A \\ A - B \neq B - A \end{array} \right\} \checkmark$$

Associative Properties:

$$\begin{array}{l} A \cup (B \cap C) = (A \cup B) \cap C \\ A \cap (B \cup C) = (A \cap B) \cup C \end{array}$$

$$(1+2)+3 = 1+(2+3)$$

Distributive Property:

$$\begin{array}{l} A \cap (B \cup C) = (A \cap B) \cup (A \cap C) \\ A \cup (B \cap C) = (A \cup B) \cap (A \cup C) \end{array}$$

$$\begin{array}{l} A \cup A = A \\ A \cap A = A \\ A \cap \bar{A} = \emptyset \\ A \cup \bar{A} = U \end{array}$$

$$\begin{array}{l} (\bar{\bar{A}}) = A \\ \bar{\emptyset} = U \\ \bar{U} = \emptyset \end{array}$$

$$\begin{array}{l} \overline{A \cup B} = \bar{A} \cap \bar{B} \\ \overline{A \cap B} = \bar{A} \cup \bar{B} \end{array}$$

De Morgan's Rules

Theorem: If A and B are finite sets, then

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Sequences: A sequence is a list of objects arranged in a definite order.

sayı = sayı

qizgileri unutma

rakıme olmalı

$$A+B = ? \quad B$$

If the list stops after $n \in \mathbb{N}$ steps, it is called a finite, if not it is called infinite

exmp: 0, 1, 0, 1, 1, 1, 0 or $[0, 1, 0, 1, 1, 1, 0]$

3, 8, 13, 18

1, 4, 16, 25

a_0, a_1, a_2, a_3

3, 8, 13, 18

starting point

$$a_0 = 3$$

$$a_n = a_{n-1} + 5$$

Recursive

Sequence

$0, 1, 4, 9, 16, 25$
 $a_0, a_1, a_2, a_3, a_4, a_5$

$$a_n = n^2, n \geq 0$$

explicit

Sequence

27.09.2024 CSE231 (Deure)

$p > 0$ absorb

$p < 0$ deliver

ideal voltage source

regardless of current

maintains constant

0 (yuvorlok)

V

• ideal sources

• dependent sources

• kirchhoff's current and voltage rule

30.09.2024 CSE211 (Ayrik)

Characteristic Function

If A is a subset of a universal set U ,

$$f_A = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$$

f_A of A is defined for each element of U

example

$U = \{\text{letters of the Eng. alphabet}\}$

$A = \{a\}$

You can detect or count the letter a in english words with this function.

example

let $U = \{1, 2, 3, 4, 5, 6\}$

$A = \{1, 2\}$

- RABIS

f_A can define 1, 1, 0, 0, 0, 0

Countable and Uncountable Sets

α Let A be a set.

If there exists a 1-to-1 and onto function from A to $\mathbb{N}$ then A is called countable.

α If A set is not countable then it is called uncountable

ex: $\mathbb{Z}$ is countable

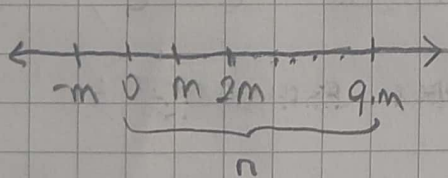
A = { the letter of Turkish alphabet }

$\mathbb{R}$ real numbers are uncountable

$$B = \{x \in \mathbb{R} \mid 0 \leq x \leq 1\}$$

Some Properties of Integers

$$n = m \cdot q \rightarrow m \mid n \quad \left(\begin{smallmatrix} m \text{ divides } n \\ n \end{smallmatrix} \right) \quad n = m \cdot q + r \quad 0 \leq r < m$$



n	m
1	9

"euclead"

if $r \neq 0$

+ then

$$m \times n$$

Theorem: Let a, b, c be non-negative integers

If $a|b$ and $a|c$, then $a|(b+c)$

proof: $b = a \cdot n$, $n \in \mathbb{Z}^+$
 $c = a \cdot m$, $m \in \mathbb{Z}^+$

$$c = Q \cdot m, \quad m \in \mathbb{Z}^+$$

$$b+c = a(n+m) \quad \xrightarrow{\quad} \in \mathbb{Z}^+$$

$$b+c = a \cdot k$$

$a \mid b+c$

02.10.2024 CSE211

* A number $p > 1$ in $\mathbb{Z}^+$ is called **prime** if the only positive integer that divides p are p itself and 1.

exmp^{le}: Is 5195731005722643 a prime number?

try $b|a$ for all b , $3 \leq b \leq \sqrt{a}$ "primality test"

Matrices: A matrix is a rectangular array of numbers arranged in m horizontal rows and n vertical columns

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \Rightarrow A \text{ is an } m \text{ by } n \text{ matrix}$$

$m \times n$ # of columns $\rightarrow$
of rows $\leftarrow$

* An array $r_1 = [a_{11}, a_{12}, \dots, a_{1n}]$ is the first row of A .

* An array $c_1 = \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix}$ is the first column of A .

$[a_{i1} \ a_{i2} \ \dots \ a_{in}]$ for $1 \leq i \leq m$ is the i th row.

$\begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}$, $1 \leq j \leq n$ is the j th column.

example:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ -1 & 0 & 1 \end{bmatrix} \rightarrow 2 \times 3 \text{ matrix}$$

$$a_{12} = 2, \quad a_{21} = -1$$

$$A = [a_{ij}] \rightarrow \text{matrix}$$

$$S = [s_i] \rightarrow \text{sequence}$$

	yas	öğrencilik durumu	Dönemi
Ali	19	1	2
Ayşe	18	1	2
Fatma	21	0	4
Ahmet	30	0	12

$= A$

$$A_{Ali, yas} = 19$$

$$A_{Ali, i} = [19 \ 1 \ 2] \quad \text{for } 1 \leq i \leq 3$$

α A - square matrix of order n has n rows and n columns.

α A square matrix $A = [a_{ij}]$ for which every entry off the main diagonal is zero, that is $a_{ij} = 0$ for $i \neq j$ is called a diagonal matrix.

* Diagonal down a 3×3 diagonal matrix

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 8 \end{bmatrix} \quad \text{main diagonal } a_{11}, a_{22}, a_{33}$$

* $a_{11}, a_{22}, \dots, a_{nn}$ are the main diagonal entries

* Two $m \times n$ matrices $A = [a_{ij}]$ and $B = [b_{ij}]$ are equal if $a_{ij} = b_{ij}$ for $1 \leq i \leq m$ and $1 \leq j \leq n$.

Sum of Two matrices

If $A = [a_{ij}]$ and $B = [b_{ij}]$ are $m \times n$ matrices, then the sum of A and B is the matrix $C = [c_{ij}]$ such that $c_{ij} = a_{ij} + b_{ij}$ for $1 \leq i \leq m$, $1 \leq j \leq n$.

example:

$$\begin{bmatrix} 1 & 0 & 1 \\ 2 & 3 & 4 \end{bmatrix} + \begin{bmatrix} 1 & -1 & -2 \\ -3 & -4 & 2 \end{bmatrix} = \begin{bmatrix} 2 & -1 & -1 \\ -1 & -1 & 6 \end{bmatrix}$$

example:

$$\begin{aligned} x - y &= 1 \\ 2x - 3y &= 5 \end{aligned}$$

matrix representation of linear system

$$\begin{bmatrix} 1 & -1 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$$

Properties of Sum of Matrices

A zero matrix $A = [a_{ij}]$ has all $a_{ij} = 0$ properties of sum of matrices

ila bogutlu matrit diklat et

- $A + B = B + A$
- $A + 0 = 0 + A = A$
- $(A + B) + C = A + (B + C)$

Multiplication of Two Matrices

If $A = [a_{ij}]$ is an $m \times p$ matrix, and $B = [b_{ij}]$ is a $p \times n$ matrix, then the product of A and B , denoted by AB is the $m \times n$ matrix $C = [c_{ij}]$ is defined by,

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{ip}b_{pj} \quad \text{for } 1 \leq i \leq m, 1 \leq j \leq n$$

example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}_{2 \times 2}$$

$$B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}_{2 \times 2}$$

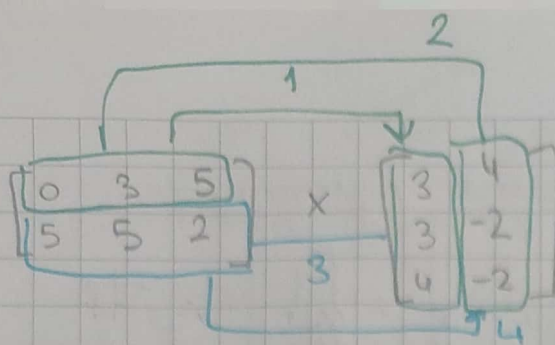
$$E \times D \neq D \times E$$

$$AB = \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix}_{1 \times 2 + 2 \times 2}$$

$$c_{11} = a_{11}b_{11} + a_{12}b_{21} = 1 \cdot 1 + 2 \cdot 3 = 7$$

$$c_{12} = a_{11}b_{12} + a_{12}b_{22} = 1 \cdot 2 + 2 \cdot 4 = 10$$

$$c_{21} = 3 \cdot 1 + 4 \cdot 3 = 15$$



- RABİS

$$\begin{bmatrix} 0 \times 3 + 3 \times 3 + 5 \times 4 & 0 \times 4 + 3 \times (-2) + 5 \times (-2) \\ 5 \times 3 + 5 \times 3 + 2 \times 4 & 5 \times 4 + 5 \times (-2) + 2 \times (-2) \end{bmatrix} = \begin{bmatrix} 29 & -16 \\ 38 & 6 \end{bmatrix}$$

$$A_{m \times n} \cdot B_{n \times k} = C_{m \times k}$$

ilk matrisin = ikinci matrisin
sütun sayısı satır sayısı

7.10.2024

1-) Give the set by listing its elements

a-) The set of all positive integers that are less than 10

$$\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

b-) $\{x \mid x \in \mathbb{Z} \text{ and } x^2 < 12\}$, $\{-3, -2, -1, 0, 1, 2, 3\}$

2-) Let $A = \{1, \{2, 3\}, 4\}$ identify each of the following as true or false

a-) $3 \in A \rightarrow F$ elemanı değil

b-) $\{1, 4\} \subseteq A \rightarrow T$ alt kümesidir

c-) $\{2, 3\} \subseteq A \rightarrow F$ $\{\{2, 3\}\} \subseteq A \rightarrow T$

d-) $\{2, 3\} \in A \rightarrow T$

e-) $\{4\} \in A \rightarrow F$ $4 \in A$

f-) $\{1, 2, 3\} \subseteq A \rightarrow F$

3- Let $A = \{1, 2, 3, 4, 5\}$ which of the following sets are equal to A ?

a) $\{4, 1, 2, 3, 5\} = A$

b) $\{2, 3, 4\} \times$

c) $\{1, 2, 3, 4, 5, 6\} \times$

d) $\{x \mid x \text{ is an integer and } x^2 \leq 25\} \times$

e) $\{x \mid x \text{ is a positive integer and } x \leq 5\} \rightarrow \checkmark$

f) $\{x \mid x \text{ is a positive rational number and } x \leq 5\} \times$

4- which of the following sets are the empty sets?

a) $\{x \mid x \text{ is a real number } x^2 - 1 = 0\} \rightarrow x^2 = 1 \rightarrow x = \pm 1$

b) $\{x \mid x \text{ is a real number } x^2 + 1 = 0\} \rightarrow x^2 = -1 \rightarrow \emptyset$

c) $\{x \mid x \text{ is a real number } x^2 = -9\} \rightarrow \emptyset$

d) $\{x \mid x \text{ is a real number } x = 2x + 1\} \rightarrow x = -1$

e) $\{x \mid x \text{ is a real number } x = x + 1\} \rightarrow \emptyset$

$\{e, \sqrt{e}\}$

5- If $A = \{3, 7\}$ find $P(A)$.

$P(A) = \{\emptyset, \{3\}, \{7\}, \{3, 7\}\}$

$|A| = 2$

$|P(A)| = 4 \quad 2^n$

6) If $P(B) = \{\{3, \{m\}, \{n\}, \{m, n\}\}\}$, B ?

$$B = \{m, n\}$$

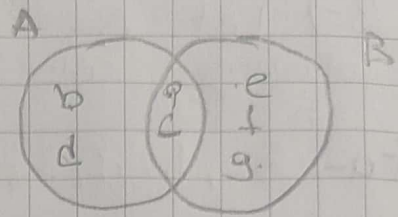
Symmetric difference:

$$A + B = (A \setminus B) \cup (B \setminus A)$$

$$A = \{a, b, c, d\}$$

$$B = \{a, c, e, f, g\}$$

$$A + B = \{b, d, e, f, g\}$$



$$7) \overline{A \cup B} = \bar{A} \cap \bar{B}$$

* De Morgan

Laws

Let's consider $x \in \overline{A \cup B}$

$$\bullet x \notin A \cup B \quad \bullet x \notin A, x \notin B$$

$$\bullet x \in \bar{A}, x \in \bar{B}$$

$$\bullet x \in \bar{A} \cap \bar{B}$$

$$\overline{A \cap B} = \bar{A} \cup \bar{B}$$

$$\bullet x \in \overline{A \cap B}$$

$$A = \{1, 2, 3\}$$

$$\bullet x \in \bar{A} \text{ or } x \in \bar{B}$$

$$B = \{2, 3, 4\}$$

$$\bullet x \in \bar{A} \cup \bar{B}$$

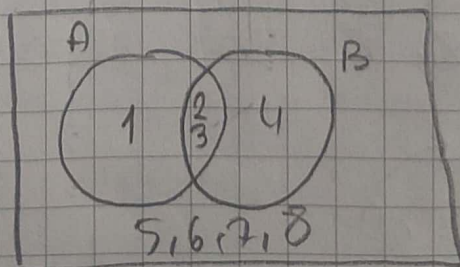
$$U = \{1, 2, 3, 4, 5, 6, 7, 8\}$$

$$\overline{A \cap B} = \bar{A} \cup \bar{B}$$

$$\{4, 5, 6, 7, 8\}$$

$$\{1, 5, 6, 7, 8\}$$

$$U \rightarrow \{1, 4, 5, 6, 7, 8\}$$



8) $U \rightarrow$ set of real numbers

$A = \{x \mid x \text{ is a solution of } x^2 - 1 = 0\}$

$$B = \{-1, 4\}$$

$$A = \{-1, 1\}$$

a) $\bar{A} = (-\infty, -1) \cup (-1, 1) \cup (1, \infty)$

b) $\bar{B} = (-\infty, -1) \cup (-1, 4) \cup (4, \infty)$

c) $\overline{A \cup B} = (-\infty, -1) \cup (-1, 1) \cup (1, 4) \cup (4, \infty)$

d) $\overline{A \cap B} = (-\infty, -1) \cup (-1, \infty)$

9.10.2024 CSE211 (Discrete Math)

Let A be a diagonal matrix with all diagonals are 1's.

A is $n \times n$ diagonal matrix such that $a_{ii} = 1$ for all $i = 0, \dots, n$

(Identity Matrix)

α

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

→ main diagonal

example:

$$\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}_{2 \times 3} \begin{bmatrix} 2 & 0 & 3 \\ 4 & 0 & 5 \\ 1 & 1 & 1 \end{bmatrix}_{3 \times 3}$$

C_{11} $0 \times 2 + 0 \times 4 + 1 \times 1$ $0 + 0 + 1$	C_{12} $0 + 0 + 1$	C_{13} $0 \times 3 + 0 \times 5 + 1 \times 1$ $0 + 0 + 1$
C_{21} $2 + 0 + 1$	C_{22} $0 + 0 + 1$	C_{23} $3 + 0 + 1$

$$\begin{bmatrix} 1 & 1 & 1 \\ 3 & 1 & 4 \end{bmatrix}$$

Let A, B, C matrices with some size.

Theorem: $A \cdot B \neq B \cdot A$ (in general)

There are some exceptions

Theorem: $(AB)C = A(BC)$

Theorem: I_n is the identity matrix size $n \times n$

$$A_{n \times n} \cdot I_n = I_n \cdot A_{n \times n} = A_{n \times n}$$

$$\begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$$

* If " $-A$ " is the additive inverse of A , then

$$A + (-A) = 0$$

* If " A^{-1} " is the multiplicative inverse of A , then

$$A \cdot A^{-1} = I$$

* we can define a power of a matrix

$$A \cdot A = A^2, \quad \underbrace{A \cdot A \cdot A \cdot A \cdot A}_{p \text{ times}} = A^p$$

* **Transpose** of a matrix $A = [a_{ij}]$ is defined by $A^T = [a_{ji}]$.

example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \Rightarrow A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

Diagram showing the mapping of elements: a_{11} to a_{11} , a_{12} to a_{21} , a_{21} to a_{12} , and a_{22} to a_{22} .

example: $0, I_3$ and all symmetric mtr hold the following

$$A^T = A$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

simetrik

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 & 3 \\ 0 & 2 & 1 & 1 & 2 \\ 0 & 1 & 3 & 2 & 3 \\ 0 & 1 & 2 & 4 & 4 \\ 1 & 2 & 3 & 4 & 5 \end{bmatrix}$$

$$A^T = A$$

-BASIS

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}_{2 \times 3} \rightarrow B^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}_{3 \times 2}$$

$$B^T \neq B$$

* If $B_{n \times n}$ is the multiplicative inverse of $A_{n \times n}$
then $A \cdot B = I_n$

NOTE: Not all square matrices have an inverse.

ex: $A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$, $A^{-1} = ?$

$$\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 \cdot a + 2 \cdot c & 1 \cdot b + 2 \cdot d \\ 2 \cdot a + 3 \cdot c & 2 \cdot b + 3 \cdot d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{aligned} a + 2c &= 1 \\ b + 2d &= 0 & b &= -2d \\ 2a + 3c &= 0 \\ 2b + 3d &= 1 & \frac{c}{2} &= 1 \end{aligned}$$

$$a = -3 \quad b = 2 \quad c = 2 \quad d = -1 \quad 2a = -3c \\ a = -\frac{3c}{2}$$

ex: $\begin{bmatrix} 0 & 2 \\ 4 & 3 \end{bmatrix}$ what is the inverse?

$$\begin{bmatrix} 3 & -2 \\ -4 & 0 \end{bmatrix} \quad \left| \begin{array}{cc|c} 0 & 2 & 1 \\ 4 & 3 & 0 \end{array} \right|$$

↳ diagonal $0 - 8 = -8$

$$\begin{bmatrix} -3/8 & 1/4 \\ 1/2 & 0 \end{bmatrix}$$

Theorem: A matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ has a unique inverse if and only if

$\boxed{ad - bc \neq 0}$ in this case,

! $A^{-1} = \begin{bmatrix} \frac{d}{ad-bc} & \frac{-b}{ad-bc} \\ \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{bmatrix} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

- LOGIC -

A statement or proposition is a declarative sentence that is either true or false, but not both.

ex: $2+3=5$ $\neq$
 $2+3=1$ $\neq$
The earth is round $\neq$
The earth is cold $-$
 $3-x=5$ $-$ (depends on x)

NOTE: In logic, the letters $p, q, r, s, \dots$ denote propositional variables. These variables can be used for statements

p : "2 is a positive number"

statements can be combined by logical connectives to obtain compound statements.

ex: p and q : 2 is a positive number and the earth is flat.

Logical Connectives:

Let p, q be true propositions

Negation: $\neg p$ or $\neg p$ or p^{-1} (olumsuzlama)

p	$\neg p$
T	F
F	T

ex: p : The water is boiling

$\neg p$: The water is not boiling

q : $2+3 < 1$

$\neg q$: $2+3 \geq 1$ veya $2+3$ is not less than 1

Conjunction: p and q , $p \wedge q$

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

ex: p : it is snowing

q : I am cold $\rightarrow$ doğru ya da yanlış olabilir
(sorgulanıyoruz)

$p \wedge q$: it is snowing and I am cold

r : $2 < 3$ s : $1 > 0$ t : $-1 > 0$

$r \wedge s$: $2 < 3$ and $1 > 0$ T

$r \wedge t$: $2 < 3$ and $-1 > 0$ F

Disjunction: p or q , $p \vee q$

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
T	T	F

ex: p : 2 is a positive number
 q : $\sqrt{2}$ is an irrational number.
 r : $2+3 \neq 5$

exclusive or: XOR: $p \oplus q$ or $p \text{ xor } q$
(yes/no)

p	q	$p \text{ xor } q$
T	T	F
T	F	T
F	T	T
F	F	F

NOTE: Instead of True or False, we use the binary variables 0 and 1, where 0 stands for "FALSE" and 1 stands for "TRUE".

Quantifiers:

when we define a set by specifying a property $P(x)$ that elements of the sets have in common.

$\{x \mid P(x)\}$, $P(x)$ is called a "predicate"
(Kosul)

ex: $A = \{x \mid x < 5 \text{ and } x \in \mathbb{Z}\}$
 $P(x)$

ex: If $\boxed{x < 5 \text{ and } x \in \mathbb{Z}}$ then x is in A
 $\rightarrow$ guard

Conditional Statements:

α If p and q are statements, the compound statement "If p then q ", denoted $p \Rightarrow q$ or $p \rightarrow q$ is called a conditional statement or implication.

$p \Rightarrow q$ (If p then q or p implies q)

hypothesis $\rightarrow$ conclusion

ex: If the degree of a polynomial is 2 then it is quadratic

(korsiti)

$q \Rightarrow p$: converse of $p \Rightarrow q$

$\neg q \Rightarrow \neg p$: contrapositive of $p \Rightarrow q$
(korsit ters)

p	q	$p \Rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

$$p \Rightarrow q \equiv q' \Rightarrow p'$$

same values

p	q	$p \Rightarrow q$	$q \Rightarrow p$	$\neg q$	$\neg p$	$\neg q \Rightarrow \neg p$
T	T	T	T	F	F	T
T	F	F	T	T	F	F
F	T	T	F	F	T	T
F	F	T	T	T	T	T

α If p and q are statements, the compound statement "p if and only if q" denoted by $p \Leftrightarrow q$ or $p \leftrightarrow q$ is called **equivalence**.

p	q	$p \Leftrightarrow q$	$(p \Rightarrow q) \wedge (q \Rightarrow p)$
T	T	T	T
T	F	F	F
F	T	F	F
F	F	T	T

$$(p \Leftrightarrow q) \equiv (p \Rightarrow q) \wedge (q \Rightarrow p)$$

$$(p \Leftrightarrow q)' = (p' \Leftrightarrow q') = (p \Rightarrow q')$$

$P \Rightarrow Q$	$Q \Rightarrow P$	$(P \Rightarrow Q) \wedge (Q \Rightarrow P)$
T	T	T
F	T	F
T	F	F
T	T	T

$\equiv \rightarrow$ equivalent

P	Q	$P \vee Q$	$Q \vee P$	$(P \vee Q) \Leftrightarrow (Q \vee P)$
				T
				T
				T
				T

$\rightarrow$ "tautology"
all of them are true

$$P \Rightarrow Q \equiv \neg P \vee Q \quad *$$

P	Q	$P \Rightarrow Q$	$\neg P$	$\neg P \vee Q$
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

Properties of Logic Operations

(dégénération)

commutativity:

$$P \vee Q \equiv Q \vee P$$

$$P \wedge Q \equiv Q \wedge P$$

(associativité)

associativity:

$$P \vee (Q \vee R) \equiv (P \vee Q) \vee R$$

$$P \wedge (Q \wedge R) \equiv (P \wedge Q) \wedge R$$

distribution: (distributivité)

$$P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$$

$$P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$$

$$P \vee P \equiv P \quad \text{and} \quad P \wedge P \equiv P$$

$$* \quad \neg(\neg P) \equiv P$$

$$* \quad \neg(P \vee Q) \equiv \neg P \wedge \neg Q \rightarrow \text{De Morgan's Rule}$$

$$* \quad \neg(P \wedge Q) \equiv \neg P \vee \neg Q \rightarrow$$

Theorem:

$$p \Rightarrow q \equiv (\neg p) \vee q$$

$$p \Rightarrow q \equiv (\neg q) \Rightarrow (\neg p)$$

$$\neg(p \Rightarrow q) \equiv ?$$

$$\neg(p \Rightarrow q) = \neg((\neg p) \vee q)$$

$$= p \wedge \neg q$$

Theorem:

$$P: \{x \mid P(x)\}$$

↳ Predicate

$$\neg(T) \equiv F$$

$$(a) \neg(\forall x P(x)) \equiv \exists x \neg P(x)$$

↓

for all

↓

there exists

$$\neg(\underbrace{\text{All even integers}}_{\forall x} \underbrace{\text{can be divided by 2}}_{P(x)})$$

↳ $\exists x$ that cannot be divided by 2
↳ even integer

$$(b) \neg(\exists x P(x)) \equiv \forall x (\neg P(x))$$

$$(c) \neg(\exists x (P(x) \Rightarrow Q(x))) \equiv \forall x P(x) \wedge \neg Q(x)$$

$$(d) \neg(\exists x (P(x) \vee Q(x))) \equiv \forall x \neg(P(x) \vee Q(x))$$

$$\forall x (\neg(P(x) \vee Q(x))) \equiv \forall x (\neg P(x) \wedge \neg Q(x))$$

21.10.2024

1- $(A^T)^T = A$

$(A \cdot B)^T = B^T \cdot A^T$

$(A+B)^T = A^T + B^T$

2- Idempotent Properties

$P \vee P \equiv P$

$P \wedge P \equiv P$

3- $\sim (P \leftrightarrow Q) \equiv (P \wedge \sim Q) \vee (Q \wedge \sim P)$

Matrix multiplication

$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$

$B = \begin{bmatrix} 7 & 10 \\ 8 & 11 \\ 9 & 12 \end{bmatrix}$

$A \cdot B = \begin{bmatrix} 1 \times 7 + 2 \times 8 + 3 \times 9 & 1 \times 10 + 2 \times 11 + 3 \times 12 \\ 4 \times 7 + 5 \times 8 + 6 \times 9 & 4 \times 10 + 5 \times 11 + 6 \times 12 \end{bmatrix}$

ex!

$P \wedge Q \Rightarrow P$

P	Q	$P \wedge Q$	$P \wedge Q \Rightarrow P$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	T

→ tautology

ex!

$P \Rightarrow (P \vee Q)$

P	Q	$P \vee Q$	$P \Rightarrow (P \vee Q)$
T	T	T	T
T	F	T	T
F	T	T	T
F	F	F	T

→ tautology

ex:

$$p \wedge q \Rightarrow q$$

* kanit gerelde truth table ile olmaz, böyle yapılır.

$$\equiv \neg(p \wedge q) \vee q$$

⇨ sinavda yaz

$$\equiv (\neg p \vee \neg q) \vee q \text{ (de Morgan)}$$

$$\equiv \neg p \vee (\neg q \vee q) \text{ (associative)}$$

$$\equiv \neg p \vee T \equiv T$$

ex:

$$q \Rightarrow (p \vee q)$$

p	q	$p \vee q$	$q \Rightarrow (p \vee q)$
T	T	T	T
T	F	T	T
F	T	T	T
F	F	F	T

→ tautology

$$q \Rightarrow (p \vee q)$$

$$\equiv \neg q \vee (p \vee q)$$

$$\equiv \neg q \vee q \vee p \text{ (commutativity)}$$

$$\equiv (\neg q \vee q) \vee p$$

$$\equiv T \vee p$$

$$\equiv T$$

} Proof

$$((p \Rightarrow q) \wedge (q \Rightarrow r)) \Rightarrow (p \Rightarrow r)$$

p	q	r	$p \Rightarrow q$	$q \Rightarrow r$	$(p \Rightarrow q) \wedge (q \Rightarrow r)$	$p \Rightarrow r$	$((p \Rightarrow q) \wedge (q \Rightarrow r)) \Rightarrow (p \Rightarrow r)$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	T
T	F	T	F	T	F	T	T
F	T	T	T	T	T	T	T
T	F	F	F	T	F	F	T
F	T	F	T	F	F	T	T
F	F	T	T	T	T	T	T
F	F	F	T	T	T	T	T

ex:

If $p \Rightarrow q$ is false, can you determine the truth value of $(\neg(p \wedge q)) \Rightarrow q$?

$$p = T \quad q = F$$

* sinavda soru sözelse
açıklamayı sözle yapmalısın
böyle değil

$$\equiv (\neg(p \wedge q)) \Rightarrow q$$

$$\equiv (\neg(T \wedge F)) \Rightarrow F$$

$$\equiv (\neg(F)) \Rightarrow F$$

$$\equiv T \Rightarrow F$$

$$\equiv F$$

ex:

If $p \Rightarrow q$ is false, $(\neg p) \vee (p \leftrightarrow q)$

$$* p \equiv T, q \equiv F$$

$$\equiv (\neg p) \vee (p \leftrightarrow q)$$

$$\equiv (\neg T) \vee (T \leftrightarrow F)$$

$$\equiv (F \vee F)$$

$$\equiv F$$

ex:

If $p \Rightarrow q$ is true, $(p \wedge q) \Rightarrow \neg q$

$\hookrightarrow p$ is true and q is true

$\hookrightarrow p$ is false and (q either true or false)

$$• p \equiv T \wedge q \equiv T$$

$$(p \wedge q) \Rightarrow \neg q$$

$$\equiv (p \wedge q) \Rightarrow \neg q$$

$$\equiv (T \wedge T) \Rightarrow \neg T$$

$$\equiv T \Rightarrow F$$

$$\equiv F$$

Theorem: Each of the followings are tautology.

i-) $(p \wedge q) \Rightarrow q$

ii-) $(p \wedge q) \Rightarrow p$

iii-) $((p \Rightarrow q) \wedge (q \wedge r)) \Rightarrow (p \Rightarrow r)$

Methods of Proof:

23/10/2024

$x: P(x)$

Direct Approach
→

If " $p \Rightarrow q$ " is tautology
 q follows p .

$(p_1 \wedge p_2 \wedge \dots \wedge p_n) \Rightarrow q$
 q logically follows $p_1, p_2, \dots, p_n$.

$\begin{array}{l} p_1 \\ p_2 \\ p_3 \\ \vdots \\ p_n \end{array}$ } hypothesis

 $\therefore q \rightarrow$ conclusion

ex: (from the previous theorem (iii))

$$\begin{array}{l} p \Rightarrow q \\ q \Rightarrow r \end{array}$$

$\therefore p \Rightarrow r$

ex: the tautology

$$(p \Leftrightarrow q) \Leftrightarrow ((p \Rightarrow q) \wedge (q \Rightarrow p))$$

$$\begin{array}{l} p \Rightarrow q \\ q \Rightarrow p \end{array}$$

$\therefore p \Leftrightarrow q$

ex: If you invest the stock market, then you'll get rich.

If you get rich, then you will be happy.

$$\begin{array}{l} p \Rightarrow q \\ q \Rightarrow r \\ \hline \therefore p \Rightarrow r \end{array}$$

$\therefore \Rightarrow$ sonua olarak onlamuna gelir.

ex:

$$\begin{array}{l} p \\ p \Rightarrow q \\ \hline \therefore q \end{array} \quad \begin{array}{l} p \Rightarrow q \\ q \\ \hline \therefore p \end{array}$$

ex: Smoking is healthy
if smoking is healthy, then cigarettes are prescribed by doctors.

$\therefore$ cigarettes are prescribed by doctors.

$$\begin{array}{l} p \\ p \Rightarrow q \\ \hline \therefore q \end{array}$$

Indirect Approach

Mostly follows from the tautology

$$(p \Rightarrow q) \Leftrightarrow ((\neg q) \Rightarrow (\neg p))$$

example: let n be an integer. Prove that

if n^2 is odd then n is odd.

$p: n^2$ is odd $\rightarrow \neg p: n^2$ is not odd

$q: n$ is odd $\rightarrow \neg q: n$ is not odd (i.e.)

$(p \Rightarrow q)$

even

$\rightarrow n=2k, \exists k \in \mathbb{Z}$

proof

If we use $(*)$, then we will show that

$(\neg q) \Rightarrow (\neg p)$ holds.

$\rightarrow$ dikkat et di'zgin ya?.

n is even $\Rightarrow n=2k \quad \exists k \in \mathbb{Z}$

$\Rightarrow n \cdot n = n \cdot 2k$

$\Rightarrow n^2 = 2nk \quad nk \in \mathbb{Z}$ since

$\Rightarrow \underbrace{n^2 \text{ is even}}_{NP} \quad k \text{ and } n \in \mathbb{Z}$

example: Prove that $\sqrt{2}$ is irrational.

$(\nexists \frac{p}{q} \in \mathbb{R} \text{ such that } (\frac{p}{q})^2 = 2) \quad \nexists x: P(x) \Rightarrow \forall x: \neg P(x)$

$n \in \mathbb{Z}$

$r \in \mathbb{Q} \quad r = \frac{p}{q}, p, q \in \mathbb{Z}$

$\frac{p}{q} \in \mathbb{R} \quad (\frac{p}{q})^2 = 2$

$\rightarrow$ negation of this one

$(p \Rightarrow q)$ There is no rational number such that its 2nd power is 2

$(\neg q \Rightarrow \neg p)$ for all rational numbers their 2nd power is not 2

$$\rightarrow \frac{1}{2} - \frac{2}{4} = \frac{3}{6} \dots$$

⊛ Let $\frac{p}{q} \in \mathbb{Q}$ such that $\gcd(p, q) = 1$

Assume, $\exists n \in \mathbb{R}$ such that $n^2 = 2$

$$n = \frac{p}{q} \Rightarrow n^2 = \left(\frac{p}{q}\right)^2 \text{ by the assumption}$$

$$\Rightarrow 2 = \frac{p^2}{q^2}$$

$$\Rightarrow p^2 = 2q^2$$

$$\Rightarrow p^2 \text{ is even}$$

① $\Rightarrow p$ is even (from the previous theorem)

$$\Rightarrow p = 2k, k \in \mathbb{Z}$$

$$\Rightarrow n^2 = \frac{p^2}{q^2}$$

$$\Rightarrow 2 = \frac{(2k)^2}{q^2}$$

$$\Rightarrow q^2 = 2k^2$$

$$\Rightarrow q^2 \text{ is even}$$

② $\Rightarrow q$ is even

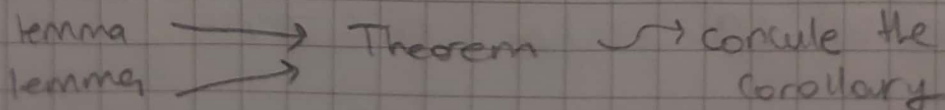
By 1 and 2 p and q are even numbers

$$\Rightarrow \gcd(p, q) = 2 \neq 1 \quad (\text{contradiction})$$

Theorem: True propositions

Lemma: A preliminary proposition useful to prove other propositions

Corollary: A proposition that follows from a theorem



Proposition: For every nonnegative integer

define $\leftarrow n^2 + n + 41$ is a prime

$$\text{Let } p(n) \equiv n^2 + n + 41$$

$$p(0) = 41 \text{ prime}$$

$$p(1) = \dots \text{ prime}$$

$\vdots$

$$p(40) = 40 \cdot 40 + 40 + 41 = 41(\dots) \text{ not prime}$$

→ A counter example

$\therefore$ By the counter example the proposition is false.

(1763) Euler's Conjecture → Euler claimed this statement to be true without a proof.

The equation;

$a^4 + b^4 + c^4 = d^4$ has no solution when a, b, c, d are positive integers.

* a counter example found 218 years later for:

$$a = 95800$$

$$b = 217513$$

$$c = 414530$$

$$d = 422481$$

NOTE! A counter example is enough to disprove a proposition.

Fermat's Last Theorem

There are no positive integers a, b, c such that

$$a^n + b^n = c^n \quad \text{for } n \in \mathbb{Z}, n \geq 2$$

Goldbach Conjecture

Every even integer greater than 2 is the sum of two primes

Correct up to 10^{18} , but don't have a proof.

Example! $\forall a, b \in \mathbb{Z}$ $a \equiv b \pmod{6}$
if and only if $a \equiv b \pmod{2}$ and $a \equiv b \pmod{3}$

Proof! we will show $(\Rightarrow)$ and $(\Leftarrow)$ are true

$(\Rightarrow)$ Let $a \equiv b \pmod{6} \Rightarrow a - b \equiv 0 \pmod{6}$
 $\Rightarrow (6 \mid a - b) \Rightarrow a - b = 6k$ for some $k \in \mathbb{Z}$
 $\Rightarrow a - b = 2(3k) = 3(2k) \Rightarrow 2 \mid a - b$ and $3 \mid a - b$
 $\Rightarrow a \equiv b \pmod{2}$ and $a \equiv b \pmod{3}$

$(\Leftarrow)$ Let $a \equiv b \pmod{2}$ and $a \equiv b \pmod{3}$

$\Rightarrow 2 \mid a - b$ and $3 \mid a - b$

$\Rightarrow a - b = 2k_1$ and $a - b = 3k_2$ for some $k_1, k_2 \in \mathbb{Z}$

$\Rightarrow$ If $2k_1$ is even then $3k_2$ is also even

$\Rightarrow a - b = 3(2k_3) \Rightarrow 6 \mid a - b$ for some $k_3 \in \mathbb{Z}$

$\Rightarrow a \equiv b \pmod{6}$

To prove iff statement:

$P \Leftrightarrow Q$

way 1 $\begin{matrix} \hookrightarrow \text{part 1} & P \Rightarrow Q \\ \hookrightarrow \text{part 2} & Q \Rightarrow P \end{matrix}$

way 2 $\rightarrow$ direct approach

works on non-empty
and countable sets

Mathematical Induction

we want to prove $P(n)$ is true for
 $n \geq n_0$, where n_0 is some fixed integer
(given)

1-) $P(n_0)$ is true (Basis step)

2-) If $P(k)$ is true for some $k \geq n_0$ then $P(k+1)$
must be true. → assumption

Then we can conclude 3-) $P(n)$ is true for all $n \geq n_0$.

örnek:

$$1+2+\dots+n = \frac{n(n+1)}{2} \quad \text{for } n \geq 1$$

let's use the mathematical induction.

i-) Basis step

$$n_0=1 \Rightarrow 1 = \frac{1 \cdot (1+1)}{2} = 1 \quad \checkmark$$

ii-) Assume for some k , we want to see

$$P(k): 1+2+\dots+k = \frac{k(k+1)}{2} \quad \left\{ \begin{array}{l} P(k+1): 1+2+\dots+(k+1) = \frac{(k+1)(k+2)}{2} \end{array} \right.$$

$$\text{we have: } 1+2+\dots+k = \frac{k(k+1)}{2} \quad +k+1$$

$$\begin{aligned} 1+2+\dots+k+k+1 &= \frac{k(k+1)}{2} + k+1 = \frac{k(k+1) + 2(k+1)}{2} \\ &= \frac{(k+1)(k+2)}{2} \end{aligned}$$

iii-) Then we can conclude that

$$1+2+\dots+n = \frac{n(n+1)}{2} \text{ is true for all } n \geq 1$$

Örnek:

Show that $n! \geq 2^{n-1}$ for $n \geq 1$

1-) $n=1 \quad 1! \geq 2^0 \quad 1=1 \quad \checkmark$

2-) assume that for some $k \geq 1$

$P(k): k! \geq 2^{k-1}$ we want to see

$P(k+1): (k+1)! \geq 2^{(k-1)+1}$

$k! \geq 2^{k-1}$

$\checkmark \rightarrow$ doğru mu? evet $\checkmark$

$(k+1) \cdot k! \geq (2^{k-1}) \cdot 2$

$(k+1) \geq 2 \quad k \geq 1$

$(k+2)(k+1)k! \geq (2^{k-1}) \cdot 2 \cdot 2$

$(k+2)(k+1) \geq 4 \rightarrow$ doğru mu? evet $\checkmark$

* Bu çözüm doğru olmasa
çünkü doğruluğunu göstermek
istediğimiz şeyden başka
kullanılmamıştır.

$k^2 + 3k + 2 \geq 4$

$k(k+3) \geq 2 \quad \checkmark$

1-) for $n=1 \Rightarrow 1 \geq 2^{1-1} \Rightarrow 1=1 \quad \checkmark$

2-) Assume that for some $k \geq 1$, $k! \geq 2^{k-1}$ holds.

we want to show that $n=k+1$

$(k+1)! \geq 2^{(k+1)-1}?$

$k!(k+1)$

$2^k = 2^{k-1} \cdot 2$

$(k+1)! \geq 2^{k-1} (k+1) \geq 2^{k-1} \cdot 2$

$k! \geq 2^{k-1}$
 $k!(k+1) \geq 2^{k-1} \cdot (k+1)$
since $k+1 > 0 \quad k \geq 1$

$(k+1)! \geq 2^{k-1} (k+1) \geq 2^{k-1} \cdot 2 = 2^k$

$(k+1)! \geq 2^k$

Örnek 1

Let $A_1, A_2, \dots, A_n$ be any n sets.

Show that

$$(*) \quad \overline{\left(\bigcup_{i=1}^n A_i \right)} = \bigcap_{i=1}^n \overline{A_i} \quad (\text{Generalized De Morgan's Rule})$$

$\Downarrow$

$$\overline{(A_1 \cup A_2 \cup \dots \cup A_n)} = \overline{A_1} \cap \overline{A_2} \cap \dots \cap \overline{A_n}$$

we will induction:

$$1) \quad n=1 \quad \overline{A_1} \cap \overline{A_1}$$

$$2) \quad n=2 \quad \overline{A_1 \cup A_2} = \overline{A_1} \cap \overline{A_2} \quad \text{By De Morgan's Rule}$$

3) Assume for some $k \geq 1$

$$\overline{\left(\bigcup_{i=1}^k A_i \right)} = \bigcap_{i=1}^k \overline{A_i}$$

Want to show (WTS):

$$\overline{\left(\bigcup_{i=1}^{k+1} A_i \right)} \stackrel{?}{=} \bigcap_{i=1}^{k+1} \overline{A_i}$$

$$\begin{aligned} \overline{\left(\bigcup_{i=1}^{k+1} A_i \right)} &= \overline{(A_1 \cup A_2 \cup \dots \cup A_k \cup A_{k+1})} \\ &= \overline{A_1 \cup A_2 \dots A_k} \cap \overline{A_{k+1}} \quad (\text{De Morgan}) \end{aligned}$$

$$\text{By } (*) = \bigcap_{i=1}^k \overline{A_i} \cap \overline{A_{k+1}} = \bigcap_{i=1}^{k+1} \overline{A_i}$$

Örnek! Show that

$n^3 + 2n$ is divisible by 3 for every non-negative integer n

Let us use the math. induction

i-) Basis step

$$n_0 = 0 \quad 0 + 0 \not\equiv 0 \pmod{3} = 0 \quad \checkmark$$

ii-) Assume for some k ,

$$P(k) : k^3 + 2k = 3m \quad \text{for some } m \in \mathbb{Z}^+$$

want to show that

$$P(k+1) \quad (k+1)^3 + 2(k+1) \stackrel{?}{=} 3n \quad \text{for some } n \in \mathbb{Z}^+$$

$$(k+1)^3 + 2(k+1)$$

$$= k^3 + 3k^2 + 3k + 1 + 2k + 2$$

$$= \underbrace{k^3 + 2k}_{3m} + 3k^2 + 3k + 3$$

$$= \underbrace{k^3 + 2k}_{3m} + 3(k^2 + k + 1)$$

$$= 3m + 3(\dots)$$

$$= 3(\underbrace{m + \dots}_n) = 3n \equiv 0 \pmod{3}$$